

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. Student counseling in higher education draws on heterogeneous institutional knowledge, including relational curricula, cohort-specific tuition schedules, statutory exemption criteria, scholarship policies, and narrative administrative regulations. This setting poses a challenge for uniform retrieval and reasoning pipelines, as different forms of institutional knowledge may benefit from different representations, retrieval mechanisms, and processing strategies. We present CTU-Chat, a centralized supervisor-routed, domain-specialized multi-agent system for Vietnamese university counseling. The supervisor directs each query to one of four specialists, while each specialist is assigned an appropriate knowledge source, retrieval path, and set of tools. Relational academic information is handled through a knowledge graph; structured tuition and scholarship information is supported by dedicated lookup and calculation functions; and narrative regulations are retrieved through hybrid document search. Evaluation shows that CTU-Chat achieves 97.0% supervisor routing accuracy, while its proposed retrieval configuration reaches 0.9600 Hit@1 on the held-out test set. These findings suggest that centralized coordination with domain-specific knowledge allocation is a promising approach for heterogeneous university counseling.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) is a key higher education institution in the Mekong Delta, with nearly 49,000 undergraduate students in the 2025–2026 academic year [1]. At an institution of this scale, advisory needs span a wide range of academic and administrative areas. Relevant regulations and policies may also change

over time. Prior research on automated advisory systems includes CAAS, which supports chat- and voice-based admission counseling for information-technology programs [10], and REBot, which focuses on academic-regulation advising through dense retrieval and category-guided graph reasoning [9]. Their reported evaluations illustrate the value of task-specific retrieval and processing strategies, but remain centered on individual advisory domains. This motivates CTU-Chat as a coordinated multi-domain architecture for heterogeneous university counseling.

University counseling presents challenges not only because it covers multiple domains, but also because institutional knowledge is represented in different forms. (1) **Academic curricula** contain hierarchical structures and prerequisite relationships; (2) **tuition schedules** are organized by admission cohort and field of study; (3) **tuition exemption policies** and (4) **scholarship policies** combine eligibility conditions with specific support rates and thresholds; while (5) **general administrative regulations** are presented in lengthy narrative documents. Therefore, no single representation or retrieval method is suitable for every domain, motivating representation-specific evidence acquisition in CTU-Chat.

RAG and GraphRAG augment LLMs with external documents and knowledge graphs [8,5,3], while agentic architectures support tool invocation and multi-agent coordination [20,18,16]. However, these approaches do not by themselves determine how heterogeneous institutional evidence should be represented, routed, and processed. Moreover, known limitations in language-model arithmetic [14] motivate deterministic calculation for rule-based financial queries.

These challenges highlights three open research gaps in automated university counseling: (1) limited domain-specialized multi-agent orchestration; (2) uniform representation of heterogeneous institutional knowledge; and (3) the lack of representation-aware evidence routing. These gaps are examined in greater detail in Section 2.

The objective of this study is to develop and evaluate CTU-Chat for heterogeneous university counseling. To address the identified gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor coordinate domain-specialized agents with isolated knowledge spaces and constrained tool access for heterogeneous university counseling tasks?
- **RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations):** What differences in context and answer quality are observed between the full heterogeneous configuration, uniform retrieval baselines, and its component ablations?
- **RQ3 (Representation-Aware Retrieval Effectiveness):** How does the complete representation-aware evidence configuration compare with sparse, dense, hybrid, and reranked baselines in retrieval effectiveness, both overall and across counseling domains?

We hypothesize that reliable university counseling is better supported by a supervisor-routed multi-agent architecture in which each specialist is restricted

to domain-appropriate knowledge representations, retrieval paths, and tools. To investigate this hypothesis and the research questions above, we propose CTU-Chat, which coordinates four domain-specialized agents for academic, financial, scholarship, and general administrative inquiries within a unified framework combining graph-based retrieval, structured data access, deterministic calculation, and hybrid document retrieval.

The remainder of this paper is organized as follows. **Section 2** reviews related work and summarizes the research gaps. **Section 3** presents the CTU-Chat architecture, evidence-acquisition methods, and domain-specialized agents. **Section 4** reports the experimental setup, results, and discussion. Finally, **Section 5** concludes the paper and outlines future work.

2 Related Work

2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11], with over 85% of lexical units forming multi-syllabic compound words (e.g., “*học phí*” [tuition fee]). While domain-adapted bi-encoders like PhoBERT capture semantic paraphrasing, Vietnamese higher education decrees rely heavily on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to mitigate both semantic drift and alphanumeric recall failures.

2.2 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [15] scores documents via term frequency (tf) and inverse document frequency (IDF) over an inverted index ($k_1 = 1.5, b = 0.75$). BM25 provides reliable exact token matching for rigid alphanumeric identifiers (decision codes, cohort labels, course codes), but cannot capture semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious or hallucinated relationships, especially when narrative documents lack explicit relational structure. Therefore, this work adopts selective rather than universal graph construction.

2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.6 Educational and University Counseling Systems

Educational chatbots assist in advising and learning support [17], with recent systems like REBot [9] combining text retrieval with category-guided GraphRAG for academic regulations. CTU-CHAT departs from prior advisory systems across three key aspects: (1) *Task Scope*: while representative prior advisory systems often emphasize single-domain text-QA or focus primarily on narrative regulations, CTU-CHAT models 113 curricula as relational property graphs and incorporates deterministic arithmetic engines (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for cohort-dependent tuition, exemptions, and scholarship thresholds; (2) *Multi-Agent Routing*: rather than a monolithic classifier-guided pipeline, a LangGraph supervisor dynamically routes queries to four bounded specialists with isolated evidence spaces and asymmetric tool privileges; and (3) *Metric Incommensurability*: prior benchmarks evaluate traditional classification accuracy and binary Q&A F_1 -scores, precluding direct numerical comparison with CTU-CHAT, which evaluates end-to-end generative fidelity via Ragas and bounded multi-agent tool execution benchmarks.

2.7 Comparative Synthesis and Research Gaps

Synthesizing established retrieval and reasoning paradigms highlights clear operational trade-offs for university counseling [2,3,6,7,15,16,18,20]. While individual techniques resolve isolated needs—lexical matching for exact regulatory tokens, dense embeddings for semantic paraphrasing, and GraphRAG for relational

prerequisites—none simultaneously address the tripartite challenge of multi-domain institutional advising.

This analysis exposes three operational gaps in automated university counseling: (1) *Lack of bounded multi-agent orchestration*: monolithic pipelines and unconstrained agent meshes lack execution guardrails, frequently triggering tool mis-selection and parameter hallucination; (2) *Limitations of uniform representation*: forcing relational curricula, tabular tuition matrices, and narrative legal text into a single retrieval space can cause prerequisite fragmentation and semantic drift; and (3) *Absence of representation-aware routing*: dynamic steering mechanisms that allocate queries across isolated representation pathways while offloading statutory arithmetic to deterministic engines remain largely unaddressed in higher-education administrative counseling. These structural requirements directly motivate the centralized supervisor-routed architecture detailed in Section 3.

3 Proposed Model

This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Here, each *specialist* represents a bounded domain agent configured with dedicated prompt constraints, an assigned evidence store, and a restricted tool registry.

3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

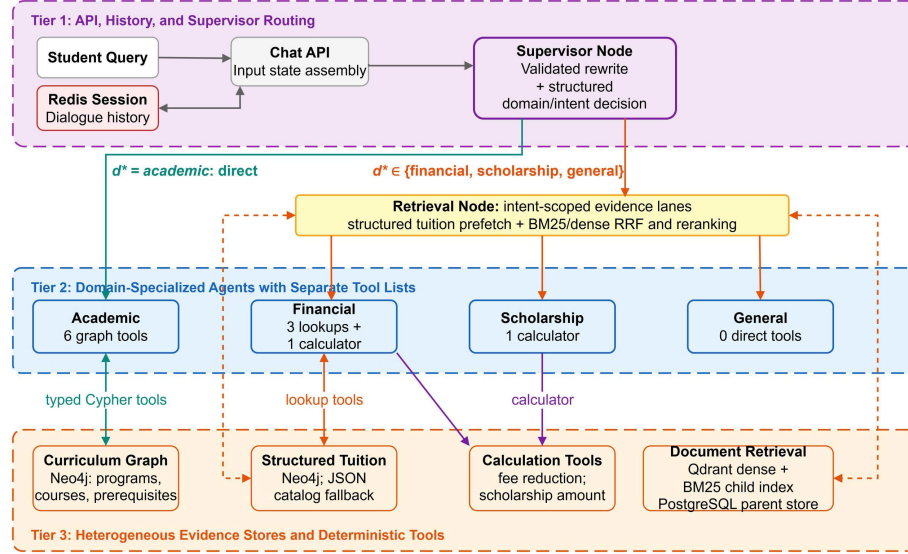


Fig. 1. Implemented CTU-CHAT architecture. The Chat API loads dialogue history from Redis and supplies it in the graph input state. The first conditional edge sends academic queries directly to the graph-enabled agent and sends the other domains through intent-scoped retrieval; the second edge dispatches the retrieved evidence to the selected domain-specialized agent. Solid agent–resource links denote direct tool access, whereas dashed links denote evidence acquisition performed by the retrieval node.

Application-Level Specialization of LangGraph StateGraph: Rather than modifying the underlying LangGraph engine, CTU-CHAT establishes an application-level orchestration policy governed by a typed state contract, isolated execution paths, and partitioned tool registries. Following supervisor dispatch, the four domain specialists operate independently without peer-to-peer delegation, selecting actions strictly from their construction-time tool bindings.

- **Node-scoped state updates:** `AgentState` defines explicit write boundaries: the supervisor writes to the routing fields (`search_query`, `next_agent`, and `routing_decision`); the retrieval node updates `context`; and each specialist returns `response`. This contract enforces transparent data ownership and mitigates cross-domain context pollution.
- **Two-Tier Conditional Routing Edges:** Instead of letting agents dynamically negotiate execution flow, the orchestration topology is governed by deterministic conditional functions:

$$\text{Edge}_{\text{sup}}(\mathcal{S}) = \begin{cases} \text{'academic_agent'}, & \text{if } \mathcal{S}[\text{'next_agent'}] = \text{'academic'} \\ \text{'retrieval'}, & \text{otherwise,} \end{cases} \quad (1)$$

bypassing document retrieval for queries assigned to the academic agent. At the retrieval exit, $\text{Edge}_{\text{ret}}(\mathcal{S})$ dispatches the retrieved context to the selected financial, scholarship, or general agent.

- **Acyclic outer orchestration:** The StateGraph has no peer-to-peer agent edges. Each selected domain-specialized agent returns to END, so the outer routing path cannot cycle. ReAct tool interactions inside an agent remain distinct from this outer graph topology.
- **Domain-specialized agents with asymmetric tools:** The academic, financial, and scholarship agents are constructed with separate prompts and tool lists, while the general agent has no direct tool (as detailed in Figure 1). This confines tool selection to the agent selected by the supervisor and keeps domain-specific evidence paths separate, reducing decision space pollution.

3.2 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

Schema defined by the ingestion layer: The Neo4j graph models relational dependencies across academic counseling using a simplified schema, comprising over 1,500 entity nodes and 3,800 relational edges across 113 undergraduate curricula and statutory fee schedules. It incorporates 10 node labels (7 for curriculum: Program, Course, Block, etc.; and 3 for tuition: TuitionFee, TuitionPolicy, ExemptionBasisRate) interconnected via 13 relationship types (HAS_BLOCK, CONTAINS, REQUIRES, PARALLEL_WITH, HAS_PLO, HAS_PEO, REALIZED_BY, HAS_TRACK, BELONGS_TO_TRACK, HAS_JOB_POSITION, HAS_TUITION, GOVERNED_BY, and HAS_EXEMPTION_BASIS). Narrative regulations remain in the document store unless linked to structured entities.

Cypher tool mechanics: The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search:* `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal:* `xem_chuoitien_quyet` follows REQUIRES paths up to ten hops and returns direct PARALLEL_WITH courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup:* parameterized queries match TuitionFee, TuitionPolicy, and ExemptionBasisRate properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. As a result, the multi-agent workflow reliably incorporates graph evidence while mitigating the syntactic and semantic fragility typically associated with unconstrained Cypher generation.

3.3 Algorithmic Foundations: Routing and Deterministic Arithmetic

Algorithm 1 specifies the supervisor-routing and evidence-path dispatch procedure.

Algorithm 1: Supervisor Routing and Conditional Evidence-Path Dispatch

Input: Student query q , Dialogue history $\mathcal{H}$, Shared state $\mathcal{S}$
Output: State updates and next graph node

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2  $q_{\text{prev}} \leftarrow \text{MostRecentUserQuery}(\mathcal{H})$ ;
3 if ShouldRewriteQuery( $q, q_{\text{prev}}$ ) then
4    $q_{\text{cand}} \leftarrow \text{FastLLM.Rewrite}(q, q_{\text{prev}})$ ;
5   if ValidateRewrite( $q, q_{\text{cand}}, q_{\text{prev}}$ ) then
6      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
7  $\langle d^*, \iota^* \rangle \leftarrow \text{StructuredSupervisor}(q)$ ;
8 if structured classification fails then
9    $d^* \leftarrow \text{'general'}$ ;  $\iota^* \leftarrow \text{'other'}$ ;
10 Update  $\mathcal{S}[\text{search\_query}] \leftarrow q_{\text{standalone}}$ ,  $\mathcal{S}[\text{next\_agent}] \leftarrow d^*$ , and
     $\mathcal{S}[\text{routing\_decision}] \leftarrow \text{QueryRoutingDecision}(\iota^*)$ ;
11 if  $d^* = \text{'academic'}$  then
12    $v_{\text{next}} \leftarrow \text{'academic\_agent'}$ ;
13 else
14   Build retrieval lanes from  $\iota^*$  and retrieve evidence;
15    $v_{\text{next}} \leftarrow d^*_{\text{agent}}$ ;
16 return  $\langle \mathcal{S}, v_{\text{next}} \rangle$ ;

```

A candidate rewrite is adopted only after validation; structured output restricts classification to declared labels with **general/other** as fallback; and conditional edges route academic queries directly to the agent while other domains pass through retrieval.

Deterministic Tuition Reduction Arithmetic: For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic tool `tinh_toan_hoc_phi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition T , the per-credit exemption basis B , and the percentage entitlement p —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (2)$$

where D is the reduction per credit and P is the remaining amount per credit. Monetary inputs are expected in VND per credit. The tool enforces $0 \leq p \leq 100$ and clamps P at zero; values outside the percentage range return an error.

Rather than a new general-purpose algorithm, the contribution lies in composing validated rewriting, constrained routing, representation-specific dispatch, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT preserves relational curriculum knowledge in Neo4j, accesses cohort-dependent tuition data through structured lookup, and retains narrative regulations and scholarship policies in document retrieval.

Table 1 makes this policy explicit by separating the physical evidence representation from the mechanism used to access it at runtime.

Table 1. Decision criteria mapping query characteristics to evidence representation mechanisms.

Query characteristic	Evidence path	Rationale
Program/course attributes	Neo4j property lookup	Exact or normalized match
Prerequisites/shared courses	Neo4j traversal	Stored graph relationships
Regulation identifiers	BM25 child index	Exact token match
Paraphrased regulations	Hybrid RRF + reranker	Lexical + semantic signals
Fee/scholarship calculation	Python tool	Fixed formula outside LLM

At runtime, the StateGraph enforces this allocation. Academic queries bypass the document-retrieval node and enter the graph-enabled academic agent directly. Financial, scholarship, and general queries enter the retrieval node, where the supervisor-assigned intent selects metadata-filtered retrieval lanes before the resulting context is passed only to the assigned domain agent.

Hybrid Document Retrieval Path. For narrative regulations and scholarship criteria, evidence is acquired through the following pipeline:

1. *Child-level candidate generation.* A BM25 index searches child chunks, while Qdrant performs dense retrieval over their 768-dimensional embeddings.
2. *Parent resolution.* Parent identifiers associated with the retrieved child chunks are used to load the corresponding parent sections from PostgreSQL.
3. *Reciprocal Rank Fusion.* The dense- and BM25-derived parent rankings are combined as

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60.$$

4. *Cross-encoder reranking.* The fused parent candidates are reranked using BAAI/bge-reranker-v2-m3, and up to six parent sections are supplied to the selected domain agent.

Parent sections use a 2,800-character target with 100-character overlap; child chunks use a 400-character target with 50-character overlap.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT.

4.1 Experimental Benchmark and Protocols

To support empirical evaluation across representative counseling scenarios, our evaluation uses a corpus of 200 administrative queries constructed via a document-grounded semi-automated protocol over official Can Tho University (CTU) statutes. Candidate queries covering diverse conversational phrasing (e.g., student colloquialisms, prerequisite rules, fee schedules) were generated from statutory articles. Each query then underwent author-guided review against official decrees to verify ground-truth intents, gold reference passages, and factual answers. Lexical and semantic duplicate checks were applied to ensure query novelty. Following standard benchmark practice, this corpus is partitioned into: (1) **Development Benchmark** ($N = 150$ queries): Spanning 9 institutional categories (`actual_tuition`, `academic_rules`, `scholarship`, `student_loan`, `other`, `social_support`, `academic_program`, `exemption_policy`, `exemption_basis`), used for offline index calibration, prompt engineering, and qualitative failure analysis; and (2) **Held-Out Test Set** ($N = 50$ queries): A separated, author-reviewed evaluation suite with exact category counts of 20, 6, 5, 4, 4, 4, 3, 2, and 2, respectively. For domain-stratified analysis, the held-out categories are grouped by assigned specialist into Academic ($N = 3$), Financial ($N = 24$), Scholarship ($N = 5$), and General ($N = 18$). All reported primary benchmark metrics in Tables 2 and 3 are evaluated on this Held-Out Test Set.

Execution and Evaluation Invariants: Generative and automated evaluation tasks were executed using Google Gemini 2.5 Flash Lite via Google Cloud Vertex AI under zero-temperature greedy decoding ($T = 0.0$). To account for subtle API-level non-determinism across cloud endpoints, each query was evaluated across $R = 3$ repeated evaluation runs, yielding 1,050 end-to-end generation and Ragas evaluation turns ($7 \text{ configurations} \times 50 \text{ queries} \times 3 \text{ runs}$).

Computational Infrastructure: Neural cross-encoder reranking was executed on an NVIDIA RTX 1060 GPU (4 GB VRAM). Relational curricula are hosted in Neo4j 5 Enterprise with APOC extensions; document parent chunks (2,800 characters, 100 overlap) reside in PostgreSQL, while child chunks (400 characters, 50 overlap) are indexed in Qdrant with `vietnamese-bi-encoder` (768-d). Neural candidate reranking uses `bge-reranker-v2-m3`. Multi-agent coordination was orchestrated via LangGraph under a StateGraph supervisor with Redis 7 caching.

4.2 Evaluation Metrics

Retrieval and Multi-Agent Benchmarks: Retrieval effectiveness (Scenario 1) is measured via standard ranking metrics on top- k returned source documents: Hit@ k ($k \in \{1, 3\}$), Precision@5, Recall@5, and Mean Reciprocal Rank (MRR@10); domain-stratified analysis reports Hit@1. Multi-agent coordination (Scenario 3) is evaluated across isolated operational decisions: Routing Agent

Accuracy and Macro-F1 across the four domains, Intent Classification Accuracy, Tool Selection and Argument Exact Match (EM), End-to-End Pass Rate, Adversarial Tool-Gating Robustness, and Safe Result Behavior under strict operational limits (`max_tool_calls` = 1, `timeout` = 60 s).

End-to-End Generative Evaluation: End-to-end synthesis (Scenario 2) is evaluated using the Ragas framework [4] alongside ground-truth regulatory alignment over Top-7 context: Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), and Answer Correctness (AC). In addition, we evaluate Factual Exact Match (Fact EM) for strict numerical tuition fee and credit agreement, alongside Source Average Precision (Source AP) for verified regulatory citations. Evaluating generative QA through LLM-as-a-judge is well-established [21,4]; to mitigate evaluation bias, our framework enforces reference-grounded metrics conditioned on gold documents, greedy decoding ($T=0.0$) across $R=3$ repeated runs, and a programmatic Fact EM evaluator for tuition arithmetic.

4.3 Experimental Scenarios and Benchmark Results

To address the three research questions, we evaluate CTU-CHAT across three coordinated experimental scenarios: retrieval component stacking (RQ3), end-to-end QA generation and modular ablation (RQ2), and multi-agent coordination and tool reliability (RQ1).

Scenario 1 — Retrieval Component Stacking (E1–E5): *Objective & RQ Alignment:* Scenario 1 directly addresses **RQ3** by comparing five retrieval configurations at aggregate and domain-stratified levels on the 50-query Held-Out Test Set. *Configurations:* We compare five cumulative configurations (E1–E5, detailed in Table 2) transitioning from single sparse/dense baselines to hybrid fusion, neural cross-reranking, and our proposed intent-guided evidence stack.

Results: As shown in Table 2, E5 attains the best Hit@1 (0.9600; 48 of 50 queries rank a gold source document first) and MRR@10 (0.9700), improving Hit@1 over E1 by 6.0 percentage points. E5 ties E4 at 0.9800 Hit@3, whereas E1 retains the highest P@5 (0.2320) and Recall@5 (0.9667). Within the shared hybrid candidate pipeline, adding the cross-encoder from E3 to E4 raises Hit@1 from 0.6800 to 0.8800 and MRR@10 from 0.7912 to 0.9267. At the domain-stratified level, E5 achieves 1.000 Hit@1 on Academic ($N = 3$) and Financial ($N = 24$), 0.800 on Scholarship ($N = 5$), and 0.944 on General ($N = 18$). The E4-to-E5 contrast improves Hit@1 and MRR@10 while leaving Hit@3, P@5, and Recall@5 unchanged; because E5 jointly adds intent governance, graph grounding, and parent-child mapping, this contrast represents the integrated bundle rather than an isolated routing effect.

Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7): *Objective & RQ Alignment:* Scenario 2 directly addresses **RQ2** by benchmarking the proposed full system (T4) against three single-agent RAG baselines (T1–T3) and three modular ablations (T5–T7) on the Held-Out Test Set

Table 2. Scenario 1: Progressive retrieval component stacking on the Held-Out Test Set ($N = 50$).

Retrieval Configuration	Hit@1	Hit@3	P@5	Recall@5	MRR@10
E1: BM25 Lexical Baseline	0.9000	0.9400	0.2320	0.9667	0.9333
E2: Dense Vector Semantic	0.3800	0.6600	0.1560	0.6433	0.5237
E3: Hybrid RRF (BM25 + Dense)	0.6800	0.9200	0.2160	0.9133	0.7912
E4: Hybrid + Neural Reranker	0.8800	0.9800	0.2200	0.9433	0.9267
E5: CTU-Chat Proposed Retrieval	0.9600	0.9800	0.2200	0.9433	0.9700

Table 3. Scenario 2: End-to-end QA generation and modular ablation on Held-Out Test Set ($N = 50$, $R = 3$, 1,050 total evaluation turns, Top-7 context).

Cfg. Ablation Variant Description	Fact EM	CP	CR	AR	AC	Source AP
T1 BM25 Lexical Baseline	0.5572	0.6674	0.8800	0.6151	0.6735	0.8954
T2 Dense Semantic Baseline	0.5007	0.2770	0.3400	0.2809	0.3841	0.4651
T3 Hybrid RRF Baseline	0.5451	0.4339	0.7733	0.5754	0.6016	0.7490
T4 CTU-Chat Full Stack (Proposed)	0.6203	0.6977	0.8933	0.6195	0.6637	0.9311
T5 w/o Cross Reranker (bge-m3)	0.5823	0.6297	0.8133	0.5263	0.6136	0.8168
T6 w/o Neo4j Knowledge Graph Grounding	0.5711	0.5683	0.8600	0.6173	0.6582	0.9011
T7 w/o Subspace Governance (Ungoverned)	0.6089	0.6947	0.8600	0.5815	0.6430	0.9311

Metrics: Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Source AP: Source Average Precision. Normalized to $[0, 1]$; higher is better.

(1,050 total evaluation turns; Table 3). *Results & Interpretation:* As reported in Table 3, T4 improves on five of six metrics relative to the BM25 baseline, with the largest gain in Fact EM (62.03% vs. 55.72%, +6.31 percentage points), while BM25 retains a marginal lead in Answer Correctness. The complementary contributions of modular cross-reranking (T5), graph grounding (T6), and subspace governance (T7) are analyzed in detail under Discussion (Section 4.4).

Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness:

Objective & RQ Alignment: Scenario 3 directly addresses **RQ1** by evaluating multi-agent coordination, tool execution reliability, and bounded safety independently from retrieval or generation. *Experimental Setup:* As reported in Table 4, evaluation spans three suites ($N = 550$ runs): (1) **Supervisor Routing (Panel A)**: 100 single-domain benchmark queries ($N = 300$ for LLM Supervisor vs. $N = 100$ for Rule-Based Router); (2) **Specialist Tool Reliability (Panel B)**: 30 test cases ($N = 90$ runs across 3 repeated runs) covering structured tuition lookup, scholarship calculation, and deterministic tuition reduction arithmetic; and (3) **Adversarial Robustness (Panel C)**: 20 failure stress tests ($N = 60$ runs across 3 repeated runs) challenging boundary invariants.

4.4 Discussion

Answering RQ1 (Domain-Specialized Multi-Agent Architecture): Centralizing routing under a structured supervisor with asymmetric privileges and

Table 4. Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per decision (`max_tool_calls` = 1).

Panel A: Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded	Routing Paradigm	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	Heuristic Regex	
LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	Structured Pydantic	
Panel B: Specialist Tool Execution Reliability ($N = 90$ runs across 30 test cases, 3 repeated runs)							
Specialist Tool Function	Runs (N)	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Paradigm
tuition_lookup	30	100.0%	100.0%	100.0%	100.0%	100.0%	Structured Catalog
scholarship.calculation	30	86.7%	76.7%	86.7%	76.7%	100.0%	LLM Tool Selection
tuition.reduction.calculation	30	100.0%	100.0%	100.0%	100.0%	100.0%	Arithmetic Tool
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate Safe	Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	–

Note: `tuition_lookup` evaluates deterministic structured catalog queries; `scholarship` and `tuition_reduction` cases evaluate LLM-driven parameter extraction and external deterministic tool execution.

bounds (`max_tool_calls` = 1, `timeout` = 60s) demonstrated reliable coordination under the evaluated configuration: the supervisor achieves 97.0% agent accuracy, 97.8% Macro-F1, and 95.0% intent accuracy across 300 runs, outperforming rule-based routing (85.0%, 69.2%). In specialist tool execution (Panel B), `tuition_lookup` and `tuition_reduction_calculation` achieved 100.0% end-to-end pass rates, while `scholarship_calculation` reached 76.7% (86.7% tool selection), with all three functions maintaining 100.0% bounded completion. Furthermore, adversarial testing (Panel C) verified 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion. To assess architectural scalability, a paired stress-test compared CTU-CHAT against a monolithic single-agent baseline: under the baseline registry (11 tools), both architectures exhibited comparable performance (single-agent 93.3% vs. multi-agent 88.3% E2E, difference not statistically significant); however, when 16 synthetic distractor tools were introduced to simulate institutional expansion (27 tools total), the monolithic agent suffered from decision-space pollution, degrading from 98.3% to 91.7% in tool selection. In contrast, domain-partitioned multi-agent maintained 100.0% tool selection (paired $\Delta = +8.3\%$, 95% bootstrap CI [+1.7, +15.0], $n = 60$). The benefit of domain-specialized multi-agent decomposition became more apparent as the tool registry expanded, because partitioned tool access reduced decision-space interference relative to the evaluated monolithic single-agent baseline.

Answering RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations): The modular ablations provide evidence that representation-specific knowledge allocation contributes to the evaluated context- and answer-quality metrics. Relative to T4, removing Neo4j (T6) is associated with a Context Precision decrease from 0.6977 to 0.5683 (18.5% relative), together with lower Factual Exact Match (62.03% to 57.11%) and Context Recall (0.8933 to 0.8600). Omitting cross-reranking (T5) is associated with lower Answer Relevancy (0.6195 to 0.5263) and Context Recall (0.8933 to 0.8133), while removing subspace governance (T7) is associated with lower Answer Correctness (0.6637 to 0.6430) and Answer

Relevancy (0.6195 to 0.5815). These comparisons support the contribution of the evaluated components within the full configuration but do not establish uniform causal effects across query types.

Answering RQ3 (Representation-Aware Retrieval Effectiveness):

The configuration sequence reveals that neural reranking principally improves early-rank ordering rather than evidence coverage: E3-to-E4 gains are concentrated in Hit@1 and MRR@10, while the full stack does not surpass the lexical baseline on P@5 or Recall@5. Domain stratification further shows that E5’s additional Hit@1 benefit is concentrated in Financial queries; General queries improve over E1 by E4 and then remain unchanged, while neither Academic nor Scholarship improves over E1. This pattern is consistent with BM25 remaining competitive on exact course codes, cohort identifiers, and administrative terminology. Because E5 changes intent governance, graph grounding, and parent–child mapping together, these results compare complete configurations and do not attribute the observed gains causally to routing alone.

4.5 Threats to Validity and Limitations

The evaluation is scoped to Can Tho University; multi-institutional validation remains planned. Domain-stratified strata are imbalanced, so small categories do not support claims of uniform cross-domain improvement. Due to budgetary constraints, Gemini 2.5 Flash Lite served as both generator and judge, introducing potential evaluation bias. Curricula graphs rely on offline schemas, and automated judges penalize valid abstentions. Fault injection across external infrastructure (Neo4j, Qdrant) is reserved for future work.

5 Conclusion and Future Work

This study investigated how centralized supervision and representation-specific evidence paths can support heterogeneous university counseling. The routing and robustness evaluations showed that the supervisor assigned queries to domain specialists with 97.0% agent accuracy while maintaining bounded execution in the evaluated cases. In retrieval, the full configuration achieved the best Hit@1 (0.9600) and MRR@10 (0.9700), whereas BM25 retained the highest P@5 and Recall@5. Domain-stratified results further showed that the additional Hit@1 benefit of the full configuration was concentrated in Financial queries rather than distributed uniformly across domains. The reranking comparison improved early-rank ordering, while the E4-to-E5 comparison reflected the performance of the integrated routing, graph-grounding, and parent–child mapping bundle rather than an isolated routing effect. End-to-end ablations were also associated with lower selected context- and answer-quality metrics when graph grounding, reranking, or governance was removed.

Taken together, these findings suggest that no single representation or retrieval strategy dominates all university-counseling tasks. Relational curricula, structured financial information, and narrative regulations impose different evidence and

reasoning requirements and are therefore better handled through separate but coordinated processing paths. In practice, the proposed architecture manages these paths within a unified deployment while restricting each specialist to its relevant knowledge sources and tools. The results support the integrated architecture in the evaluated setting, but do not establish that any individual component is universally optimal.

The evaluation remains limited to Can Tho University data, automated judging, offline knowledge schemas, small and imbalanced domain strata, and isolated robustness tests. Future work will evaluate transferability across institutions, conduct user studies with students and advisors, perform controlled computational-efficiency evaluation and model compression, employ cross-model evaluation to reduce judging bias, and compare CTU-CHAT with broader educational AI systems. Future work will also expand the production tool registry and evaluate matched single- and multi-agent configurations across increasing tool-set sizes, including production tools rather than synthetic distractors. Within the scope of this study, the central takeaway is not that one retrieval method should replace all others, but that institutional counseling is better supported when evidence acquisition and execution constraints are aligned with the structure of each domain.

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